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BIRD WATCH

The Monash Uncrewed Aerial Systems Newsletter



Issue No. 3



In This Month's Issue...

HUMANITARIAN NEED FOR DRONES (PAGE 3)

Learn a bit about our motivation behind exploring the humanitarian purpose of UAVs.



PROJECT SPOTLIGHT: LIFELINE (PAGE 6)

Learn about our exciting Lifeline Project.

PROJECT SPOTLIGHT: PEREGRINE (PAGE 9)

Learn about our exciting Peregrine Project.



HOVER ZONE (PAGE 16)

Test your knowledge of flight, and the world with this new newsletter section, the Hover Zone.



Why do We Believe in the Humanitarian Application of Drones?

By Yasiru Peiris, Chief Editor

Drones have experienced a significant resurgence in recent years, rapidly gaining popularity across various sectors. With this surge, we have witnessed a diverse range of applications for drones, from package delivery and video recording—an area we have effectively leveraged—to even emergency blood delivery. The versatility and customisability of drones are aspects that our team strongly believes in. However, we have observed that the use of drones and UAVs for humanitarian purposes remains heavily underutilised.

Why did we choose to focus on the humanitarian application of drones? The global need for humanitarian aid is immense, with an estimated 300 million people requiring assistance. The value of the aid needed is projected to be around \$47 billion USD. Many of the regions in need are war-torn or lack the necessary infrastructure for aid providers to easily access those who require assistance. Disseminating aid to these areas is often prohibitively expensive or logistically impossible due to challenging terrain.

We are not alone in recognizing this potential. The United Nations World Food Programme has highlighted that drones could "bolster responses" and be particularly useful in "disaster-prone regions" due to their ability to "circumvent poor infrastructure." The application of drones is not merely theoretical. UNICEF has conducted extensive testing of drones in various circumstances, including in the Malawi and Kazakhstan corridors and for vaccine deliveries in Vanuatu. They have specifically noted that drone technology could have significant implications for emergencies and the delivery of medical diagnostic kits. This is where our focus lies—harnessing the capabilities of drones to meet these critical needs.



Our designs and plans for the Albatross UAV have embodied this belief in the utility of drones. A key project in achieving this goal is our lifeline system—an in-house designed and manufactured payload deployment system. This system has undergone various iterations and rigorous testing to ensure its effectiveness and accuracy. You can learn more about this project later in the newsletter.

For UAVs to successfully deploy aid, it is crucial that they do not scatter payloads over a wide area, hoping they will be found by those in need. Instead, a more effective method involves deploying payloads directly to those who require them, particularly in regions with difficult-to-navigate terrain. Our solution to this challenge is our in-house developed machine vision software, Vision. This software has been developed using machine learning, with the model trained in various scenarios to identify, locate, and guide the drone and payload to the precise location identified by the software.



In conclusion, the potential of drones in humanitarian efforts is vast and largely untapped. At Monash Uncrewed Aerial Systems (MUAS), we are committed to pushing the boundaries of what UAVs can achieve, particularly in providing critical aid to those in hard-to-reach areas. Our innovative projects, such as the lifeline system and Vision software, are testaments to our belief in the transformative power of drone technology. We hope our work not only contributes to the field but also inspires others in the industry to explore the humanitarian applications of UAVs, ultimately making a positive impact on the world.



by Yasiru Peiris

Chief Editor

Project Spotlights

In this edition of Bird Watch, we decided to take a different approach. While we enjoy showcasing the progress our sections make, it doesn't fully capture the breadth of work our team undertakes. This time, we want to highlight some of our most exciting projects that drive our mission forward.

We hope you enjoy the deeper insights into some of our key projects.



Project Spotlight

LIFELINE

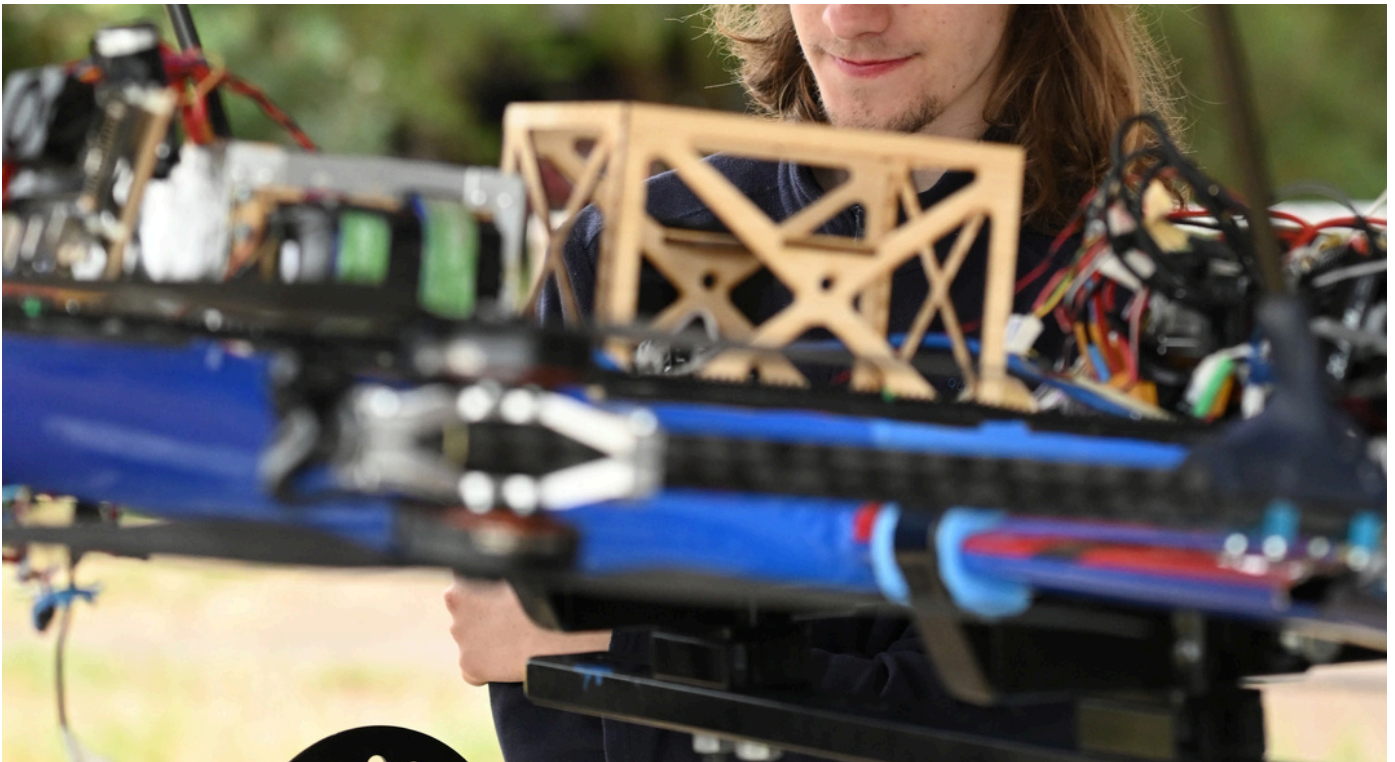


by Byran Lai

Lifeline Project Lead

What is Lifeline?

Our team has been working tirelessly on Project Lifeline, a unique multidisciplinary project that involves both aerostructure & avionics members. The project aims to develop and refine a payload deployment system that would transport a payload box, filled with essential medical supplies, safely from our humanitarian drone, the Albatross, to an injured person on the ground.



Project Vision

The project's purpose is to pioneer delivering mission critical payloads, filled with medical supplies, safely from humanitarian drones to patients in desperate need.

The development and implementation of our payload deployment system within the Albatross can demonstrate to many drone companies that drones can be used in many first aid and emergency situations, rather than just for military prowess and recreational activities. This will push the incentives to develop more humanitarian drones and deliver more first-aid payloads to people in need.

The payload deployment system is designed to fit within the compact space of the Albatross and its software is designed to be integrated with Mission Management control panel. Pairing with the latest vision system, the team aims to detect a person on the ground and trigger a signal to our deployment system, which initiates the payload winching sequence.

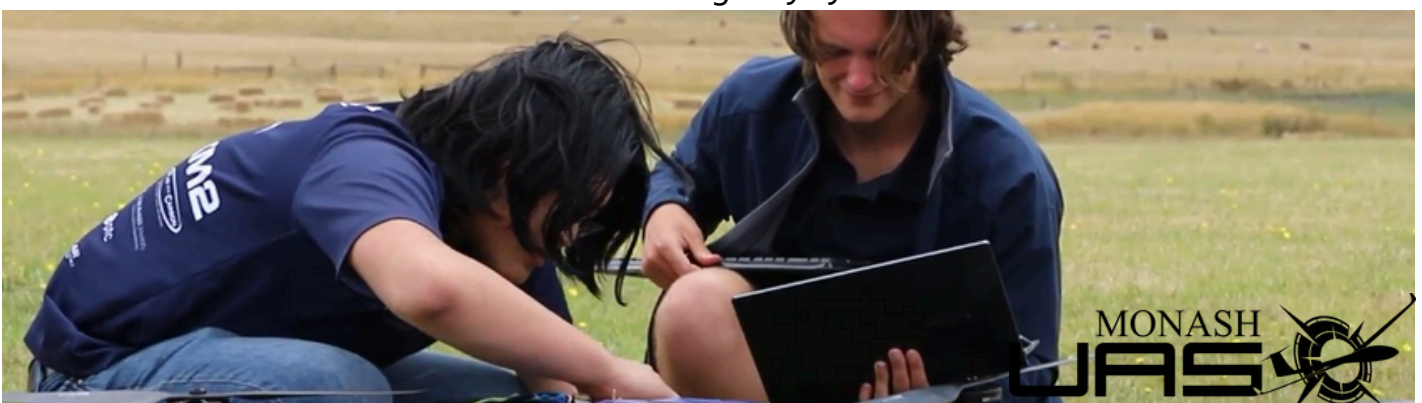
The payload would be first be released to freefall. Once the payload reached a certain distance above the ground, the brake mechanism would immediately actuate and slow the payload down. After the cables attached to the payload have been detached and the payload has been released onto the ground, the system would winch the cable back up into the Albatross, completing the sequence.



Composite Creation

Lifeline's Development

The project started in July 2023 and has come a long way since then. The first prototype included the actuation of the Albatross bottom doors and the releasing of the payload midair. It marked an important milestone, which involved the successful integration of the Lifeline system with Mission Management through MAVLINK connection and successful use of N.E.R.F emergency system.



MUAS Aerostructures Team

The N.E.R.F system is an emergency system that burns the cables and releases the payload early midair should there be any threat to the aircraft safety.

In the upcoming milestone, the system is now being upgraded to include the breathtaking brake system, which is heavily inspired by car and bicycle brake systems. The spool that is wrapped by the cables attached to the payload is also being upgraded to be easily detachable for easier re-use and the latest version of the N.E.R.F system is being incorporated



MUAS Ground Station

The seamless collaboration between our dedicated team members and various other specialized groups, including Mission Management, Vision, and Stack is integral to the development. Our team members dive deep into a diverse range of tasks, from working with intricate CAD models and establishing MAVLINK connections to designing PCBs, conducting rigorous structural testing, and programming microcontrollers.

Weekly testing sessions are a cornerstone of our process. These regular check-ins help us catch potential issues early and allow us to refine our systems continuously. Our senior members, with their wealth of experience, provide invaluable guidance, helping to steer the team through any challenges that arise.

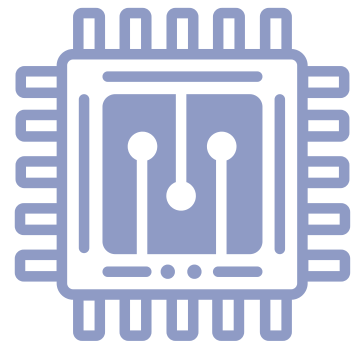


MUAS Flight Operations Team

Nichrome Emergency Release Failsafe

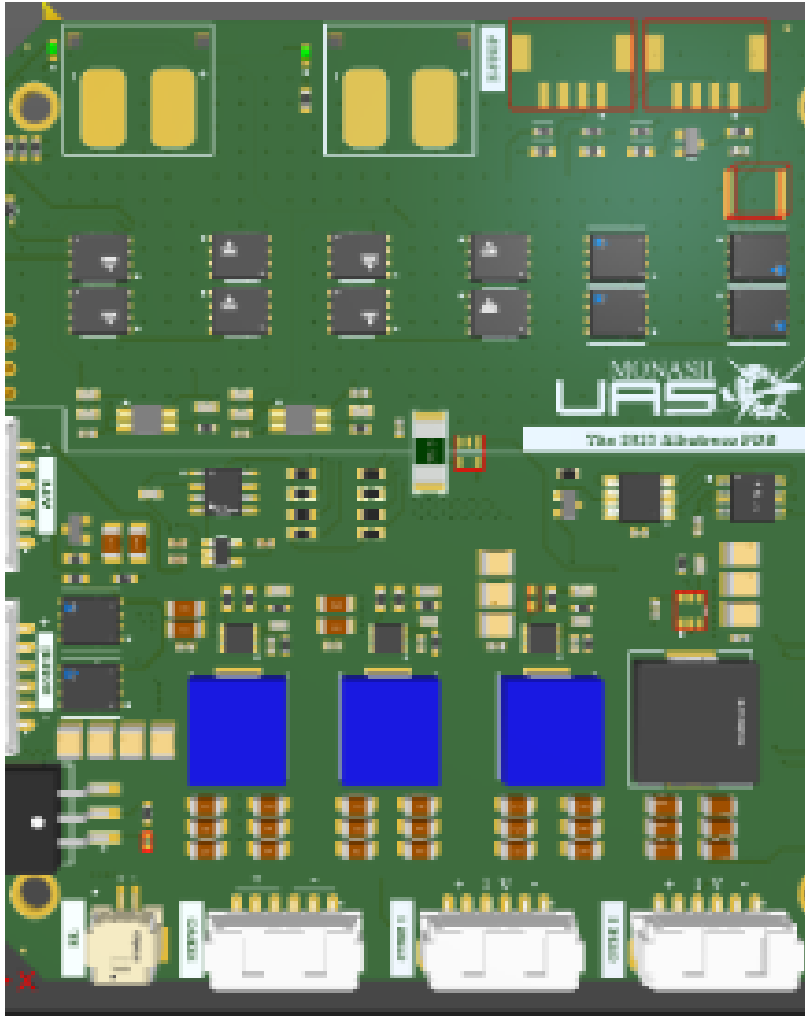
The project incorporates an emergency system, named N.E.R.F, which no other engineering firm in the world has attempted. Its acronym is based on the words “Nichrome Emergency Release Failsafe”. After over a year of continuous testing and experimenting, the system has been finally tested to be a success and has been integrated into a compact PCB for the next Milestone.

PEREGRINE



by Thomas Bugno

Aerostructures Lead



The Power Distribution Board

What is Peregrine?

We are excited to introduce our latest innovation: a scale model of our fixed-wing drone, designed specifically to test and refine the key features and technologies that will be integral to our final aircraft. This 2.3m wingspan prototype, representing two-thirds the size of the planned full-scale version, is a pivotal part of our development process. It serves as a crucial testing ground for aerodynamic efficiency, control systems, and overall design, providing invaluable insights that will guide the final iteration of the drone. The scale model's development marks a significant milestone in our journey, showcasing our commitment to precision engineering and innovative design.

Project Vision

The purpose of developing the scale model of our fixed-wing drone is to rigorously test and validate the design principles and technologies that will be scaled up in the final aircraft. This 2.3m wingspan prototype serves as a critical step in our development process, allowing us to explore and refine key aerodynamic and control features before committing to the full-scale build.

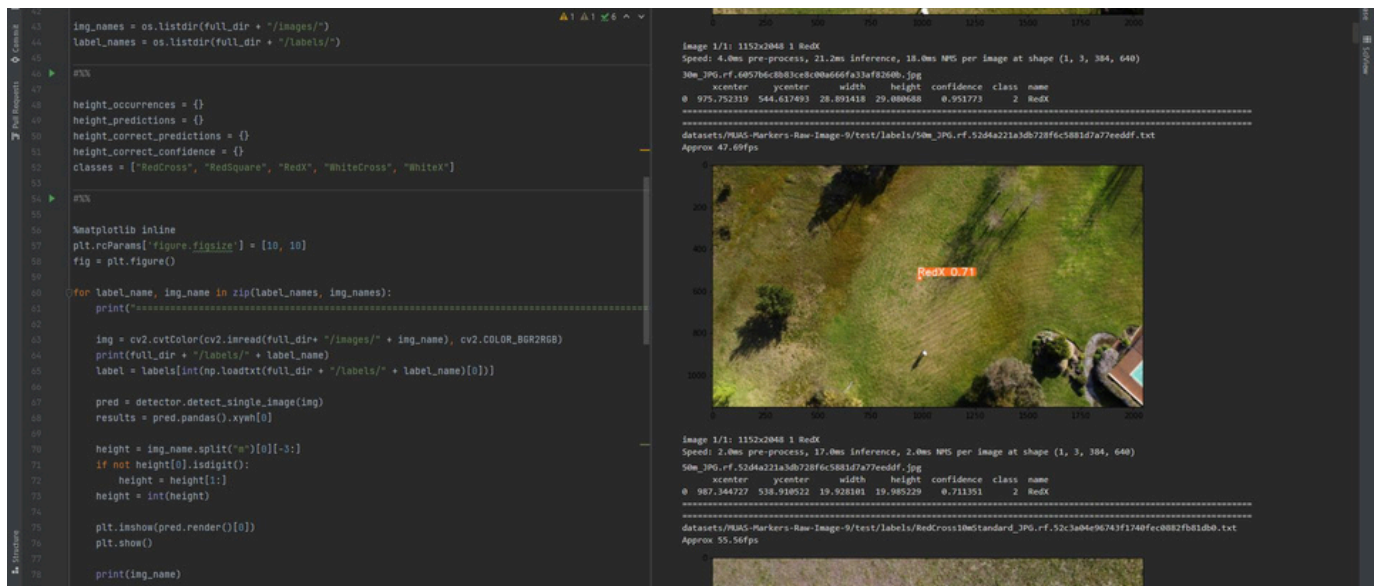
The scale model is specifically designed to test various configurations and modifications in a controlled and cost-effective manner. By conducting extensive testing on this smaller version, we can ensure that the final design will be optimised for efficiency and reliability. This iterative approach not only helps in identifying and solving potential issues early in the development cycle but also provides valuable data that informs the design and functionality of the final aircraft.



Mission Management

The focus of this project phase is on achieving controlled, stable flight, validating the aerodynamic design, and ensuring the aircraft's control systems are robust. This meticulous testing and refinement process will lay a solid foundation for the final drone, ensuring that it meets all performance and safety standards required for its intended applications, including delivering medical supplies in challenging environments.





Pipeline Programming

Peregrine’s Development

The development process for the drone has been comprehensive and methodical. The journey of our 2.3m wingspan prototype drone, which is two-thirds the size of the final 3.4m design, has been marked by meticulous planning and a series of carefully structured development phases. Each phase aims to validate different aspects of the design, ensuring that by the time we reach the NFC competition, our drone will be fully optimized for efficiency and performance.

Initially, our team focused on aerodynamic design, employing advanced simulation tools to predict airflow patterns and optimise the wing shape. The unpowered first version provided invaluable real-world data, allowing us to refine the design and address any unforeseen issues.

The progression to a powered version involved integrating a propulsion system that complements the aerodynamic characteristics of the drone. This stage also included enhancing the control systems to ensure stability and responsiveness during flight. The decision to continue using a slingshot launch method for this version allows us to isolate and test the propulsion and control systems without the added variables of take-off dynamics.

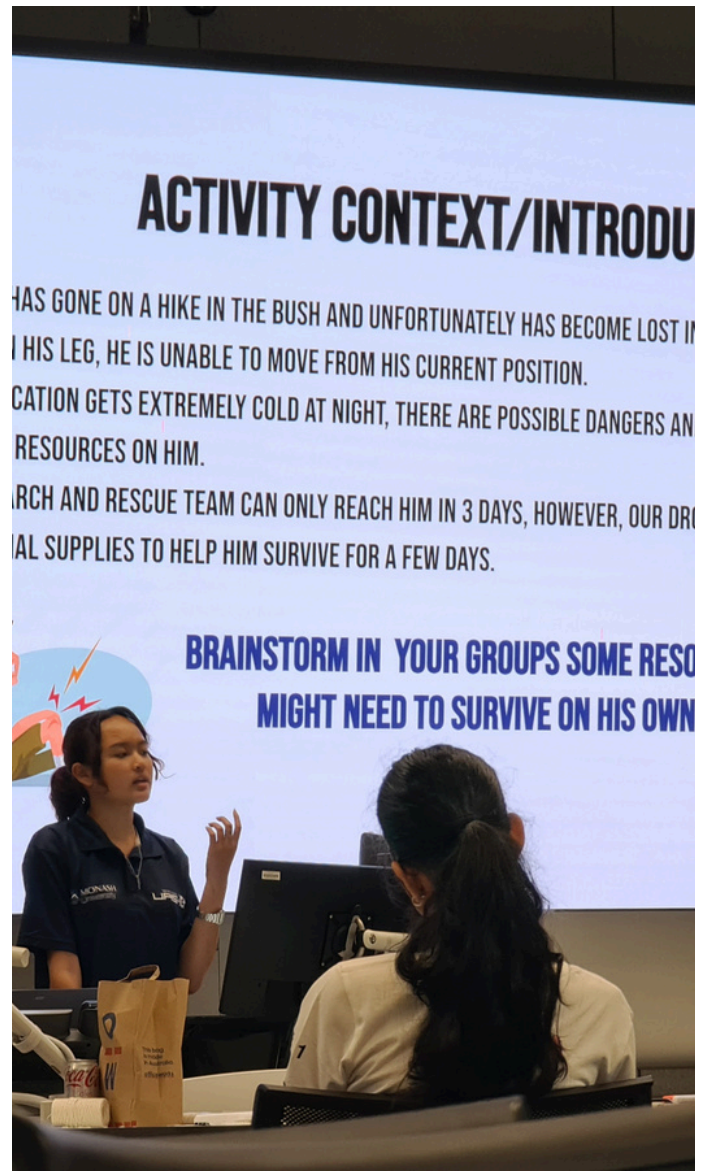
Looking ahead, the third version of the drone will include landing gear, enabling it to take off under its own power. This phase is crucial as it will test the drone's performance under ground effect, a phenomenon that can significantly alter an aircraft's control characteristics during take-off and landing.



Working towards NFC 2025

One of the most fascinating aspects of this project has been the iterative learning process. Each test flight has provided new insights, allowing us to continuously refine and improve the design. The team's adaptability and problem-solving skills have been paramount in overcoming challenges, such as initial difficulties in achieving stable flight with the unpowered version.

Another intriguing element is the emphasis on efficiency and performance in the context of the NFC competition. The competition's focus on these aspects has pushed us to innovate, exploring lightweight materials and optimizing the propulsion system for maximum efficiency. Our participation in NFC is not just about winning; it's about pushing the boundaries of drone technology and demonstrating the potential for these aircraft to perform critical tasks, like delivering medical supplies in disaster zones or remote areas



Engage Engineer Outreach Event

As we prepare for the next phases, the excitement is palpable. The upcoming powered flights and the eventual inclusion of landing gear will be critical in proving the drone's capabilities. The final goal is a fully operational, efficient, and reliable drone that meets the high standards of the NFC and beyond.



Monash Motor Sports Drone Workshop Outreach Event

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Hover Zone QUIZ

Questions

1. The Wright brothers invented flight, what were their first names?
2. In the NATO Phonetic Alphabet what word represents the letter 'K'?
3. What was the first song ever played on MTV?
4. What is the name of the world's tallest waterfall?
5. How many sides does a heptagon have?
6. What two forces are required for flight?
7. What country is the largest producer of oil?
8. According to Greek Mythology who flew too close to the sun?
9. Which planet in our solar system is the hottest?
10. Which bird has the largest wingspan?

Answers

1. Wilbur and Orville Wright
2. Kilo
3. Video Killed the Radio Star
4. Angel Falls
5. 7
6. Lift and Thrust
7. USA
8. Icarus
9. Venus
10. Albatross